

Labwork 1: Gradient Descent Report

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1 Introduction

This report presents an implementation of gradient descent from scratch to minimize the function:

$$f(x) = x^2$$

The derivative is:

$$f'(x) = 2x$$

2 Algorithm

The update rule used is:

$$x = x_A - r \cdot f'(x_A) = x - 2rx_A$$

where r is the learning rate.

3 Implementation

The algorithm is implemented in Python without using any libraries. The process iteratively updates x until convergence or until a maximum number of iterations are reached.

4 Iteration Table Example

Iteration	x	f(x)
0	5.0	25.0
1	4.0	16.0
2	3.2	10.24
3	2.56	6.553
4	2.048000	4.194304
5	1.638400	2.684355
6	1.310720	1.717987
7	1.048576	1.099512
8	0.838861	0.703687
9	0.671089	0.450360
10	0.536871	0.288230
11	0.429497	0.184467
12	0.343597	0.118059
13	0.274878	0.075558
14	0.219902	0.048357
15	0.175922	0.030949
16	0.140737	0.019807
17	0.112590	0.012677
18	0.090072	0.008113
19	0.072058	0.005192
20	0.057646	0.003323
...		
62	0.000005	0.000000

5 Effect of Learning Rate

5.1 Small Learning Rate

Convergence is slow but stable.(lr=0.05-126 epochs)

5.2 Moderate Learning Rate

Faster and more stable convergence. (lr=0.1-63 epochs)

5.3 Large Learning Rate

Fastest but may cause oscillation or divergence.(lr=0.5-2 epochs)

6 Conclusion

Gradient descent successfully converges to the minimum at $x = 0.000004$.